PNPInverse Weekly Development Writeup
Week of March 4, 2026 — Reference Document

Jake Weinstein

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Weekly Focus

This week's work focuses on a systematic **exploration of the $k_{0,1}$–$k_{0,2}$ accuracy trade-off** that has persisted across all inference pipeline versions (v7–v9). While the multi-observable master protocol (v3/v7) achieved the first-ever simultaneous recovery of all four BV parameters to within 17%, no version has achieved $< 10\%$ error on all four parameters simultaneously. The root cause is a fundamental gradient competition between the two reaction rate constants arising from their $24\times$ scale disparity and the equal weighting of observables in the least-squares objective. Seven exploration strategies are proposed and investigated: Strategies 1–6 exhaustively characterize the surrogate accuracy landscape; Strategy 7 (v11) combines surrogate warm-starting with PDE refinement to break the $< 10\%$ barrier for the first time (max err = 9.43%). An overnight $7\times$ data expansion and multi-architecture surrogate training pipeline is then launched to further improve surrogate fidelity.

1. Inference Pipeline Accuracy Exploration

1.1 Root Cause Analysis: $k_{0,1}$ vs. $k_{0,2}$ Trade-Off

The core identifiability challenge in the PNP–BV inverse problem arises from the interaction of three factors:

1.1.1 The 24× Scale Disparity

The two exchange rate constants differ by a factor of 24 in physical units:

$$k_{0,1}^{\text{phys}} = 2.4000 \times 10^{-8}\,\text{m/s} \quad (\text{O}_2 \to \text{H}_2\text{O}_2), \tag{1}$$

$$k_{0,2}^{\text{phys}} = 1.0000 \times 10^{-9}\,\text{m/s} \quad (\text{H}_2\text{O}_2 \to \text{H}_2\text{O}). \tag{2}$$

In dimensionless form (nondimensionalized by $K_{\text{scale}} = D_{\text{ref}}/L_{\text{ref}} = 1.9000 \times 10^{-5}$ m/s):

$$\hat{k}_{0,1} = K_{0,1}^{\text{phys}}/K_{\text{scale}} \approx 1.263 \times 10^{-3}, \tag{3}$$

$$\hat{k}_{0,2} = K_{0,2}^{\text{phys}}/K_{\text{scale}} \approx 5.263 \times 10^{-5}. \tag{4}$$

The ratio $\hat{k}_{0,1}/\hat{k}_{0,2} = 24.0$ means that R1 contributes ∼24× more current per unit overpotential than R2, making the total current density overwhelmingly dominated by R1 in the kinetically controlled regime.

1.1.2 Gradient Competition from Equal Observable Weighting

The multi-observable objective function is:

$$J(\theta) = \underbrace{\frac{1}{2}\sum_i \left(I_{\text{tot},i}^{\text{sim}}(\theta) - I_{\text{tot},i}^{\text{target}}\right)^2}_{J_{\text{cd}}} + w_2 \cdot \underbrace{\frac{1}{2}\sum_i \left(I_{\text{pxd},i}^{\text{sim}}(\theta) - I_{\text{pxd},i}^{\text{target}}\right)^2}_{J_{\text{pc}}}, \tag{5}$$

where $\theta = [\log_{10} k_{0,1},\ \log_{10} k_{0,2},\ \alpha_1,\ \alpha_2]$, $I_{\text{tot}} = -(R_1 + R_2) \cdot I_{\text{scale}}$ is the total current density, and $I_{\text{pxd}} = -(R_1 - R_2) \cdot I_{\text{scale}}$ is the peroxide partial current.
With equal weighting ($w_2 = 1.0$), the two terms contribute comparably to the total gradient $\nabla_\theta J$. However, the *sensitivity* of each observable to R2 parameters differs markedly:

- I_{tot} is dominated by R1 at all overpotentials (because $k_{0,1} \gg k_{0,2}$). The gradient $\partial J_{\text{cd}}/\partial \log_{10} k_{0,2}$ is therefore small, and the optimizer preferentially adjusts $k_{0,1}$ and α_1 to reduce J_{cd}.
- $I_{\text{pxd}} = -(R_1 - R_2) \cdot I_{\text{scale}}$ is also dominated by R1 in the onset/transition regime, but R2 begins to subtract from R1 at larger overpotentials (as H_2O_2 consumption accelerates). Thus J_{pc} is the *only* term with meaningful R2 sensitivity.
- When $w_2 = 1.0$, the R1-dominated gradient from J_{cd} competes with the R2-sensitive gradient from J_{pc}. The optimizer reaches a compromise that fits R1 well ($k_{0,1}$ error 7–11%) but cannot simultaneously drive $k_{0,2}$ error below ∼7–10%.

1.1.3 Voltage Range Sensitivity

The R2 contribution to both observables is concentrated in the "knee" region ($6 < |\hat{\eta}| < 12$), where the peroxide concentration at the electrode builds up enough for R2 to become non-negligible. In the deep plateau ($|\hat{\eta}| > 12$), both reactions are diffusion-limited and the I–V curve is flat, providing no additional discriminatory power. In the onset region ($|\hat{\eta}| < 2$), both reactions are exponentially suppressed and R2 is invisible.
This means:

- The "shallow" voltage grid ($\hat{\eta} \in [-1, -13]$, 10 points) captures the informative knee region but lacks plateau data.

- The "full cathodic" grid ($\hat{\eta} \in [-1, -46.5]$, 15 points) adds plateau data that tightens the limiting-current constraint but provides diminishing returns for R2.
- Symmetric (anodic + cathodic) data provides an independent constraint on α via the $(1-\alpha)$ Tafel slope, but does not resolve the $k_{0,1}$–$k_{0,2}$ amplitude trade-off.

1.2 Historical Error Summary

Table 1 summarizes the best parameter recovery results from each pipeline version. In all cases, 2% Gaussian noise was applied to synthetic targets generated at the true parameter values.

Table 1: Historical parameter recovery errors (%) across pipeline versions. Best result per version (lowest max error across all 4 parameters). Target generation (∼70 s) is excluded from optimization time.

Version	Phase	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Optim. time
v3 (PDE, serial)	P3	16.7	6.1	8.2	11.7	16.7	∼2075 s
v7 (PDE, parallel)	P3	10.9	2.6	5.6	8.8	10.9	∼345 s
v8 (Surrogate, 181)	P2	7.3	9.6	4.5	4.6	9.6	< 0.1 s
v9 (Surrogate, 445)	P2	8.8	7.6	4.8	6.4	8.8	< 0.1 s

Key observations from the historical data:

1. **R1–R2 trade-off:** v7 (PDE) achieves the best $k_{0,2}$ (2.6%) but the worst $k_{0,1}$ (10.9%) of the competitive methods. The surrogates achieve better $k_{0,1}$ (7.3–8.8%) at the cost of degraded $k_{0,2}$ (7.6–9.6%).
2. **Alpha accuracy is consistently good:** All versions achieve $\alpha_1 < 9\%$ and $\alpha_2 < 12\%$. The trade-off is almost entirely between $k_{0,1}$ and $k_{0,2}$.
3. **No version achieves $< 10\%$ on all four parameters simultaneously.** The v9 surrogate comes closest with max error 8.8% (all parameters below 9%), but this required 445 training samples (∼9.5 hours offline).
4. **Diminishing returns from more training data:** v8 (181 samples) → v9 (445 samples) improved $k_{0,2}$ by only 2 percentage points (9.6% → 7.6%) at 3× training cost.

1.3 Exploration Strategies Overview

Seven strategies were proposed to break the $k_{0,1}$–$k_{0,2}$ accuracy ceiling. All seven were investigated this week.

1. **Strategy 1: Adaptive Observable Weighting** (§). Sweep the weight w_2 on the peroxide current term J_{pc} in the objective. **Result:** $w_2 = 1.0$ is already minimax-optimal (8.76%).

2. **Strategy 2: Block Coordinate Descent (BCD)**. Alternate between optimizing R1 and R2 parameter blocks. **Result:** max error worsened to 22.10% due to flat valley on full voltage grid.

3. **Strategy 3: Multi-Start LHS Grid Search**. 20,000 Latin Hypercube initial points + local polishing. **Result:** all 20 polished candidates converge to the same 8.76% optimum, confirming it is the global minimum.

4. **Strategy 4: Per-Observable Cascade** (§). Use $w_2 = 0.5$ to lock in $k_{0,1}$ at ∼0.51%, then fix R1 parameters and optimize $k_{0,2}$ in a reduced 2D search. **Result:** 3/4 parameters excellent (max 2.38%), but $k_{0,2} = 16.60\%$; joint polish erases cascade gains.

5. **Strategy 5: Fixed PDE Refinement** (§). Re-enable PDE refinement with safeguards (tight bounds, regularization, early stopping). **Result:** max error = 8.73% (tiny improvement); PDE no longer regresses.

6. **Strategy 6: Cascade + PDE Hybrid** (§). Use CD-dominant cascade to lock in 3/4 parameters, then PDE refinement on $k_{0,2}$. Four sub-iterations investigated: (6a) original 1D PDE, 14.91%; (6b) multi-observable PDE, 12.51%; (6c) asymmetric regularization, 12.51%; (6d) free joint PDE from cascade warm start, 13.60% (Pareto tradeoff). **Root cause:** cascade's small errors in 3 params bias the PDE-optimal $k_{0,2}$ by ∼12% due to tight 4-parameter coupling; freeing all 4 reveals a fundamental $k_{0,1}$–$k_{0,2}$ anticorrelation.

7. **Strategy 7: Surrogate-Warm-Started PDE Pipeline (v11)** (§). Replace v7's expensive cold-start PDE phases with the v9 surrogate for warm-starting, then run v7's proven PDE protocol with wide bounds, no regularization, and the full cathodic grid. **Result:** max err = 9.43%; **all four parameters** $< 10\%$ **simultaneously for the first time**. $k_{0,2} = 0.82\%$ (best ever). 30% PDE wall-time reduction.

2. Strategy 1: Adaptive Observable Weighting

2.1 Motivation

The simplest intervention is to increase the secondary weight w_2 in Equation (5). With $w_2 > 1$, the optimizer places more emphasis on fitting the peroxide current, which is the only observable with meaningful sensitivity to R2. This amplifies the $\partial J/\partial k_{0,2}$ gradient component relative to $\partial J/\partial k_{0,1}$, potentially allowing the optimizer to find a better compromise.

2.2 Implementation

The existing `SurrogateObjective` class in `Surrogate/objectives.py` already accepts a `secondary_weight` parameter (default 1.0). The objective computation is:

```
J = 0.5 * sum((cd_sim - cd_target)**2)
    + secondary_weight * 0.5 * sum((pc_sim - pc_target)**2)
```

The v9 surrogate pipeline (`scripts/surrogate/Infer_BVMaster_charged_v9_surrogate.py`) uses $w_2 = 1.0$ in both Phase 1 (alpha-only) and Phase 2 (joint 4-parameter). Changing w_2 requires no code modifications beyond passing a different value to the `SurrogateObjective` and `SubsetSurrogateObjective` constructors.

2.3 Weight Sweep Design

A systematic sweep over $w_2 \in \{0.1, 0.25, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$ was conducted using the v9 surrogate pipeline. For each weight:

1. Phase 1: Alpha-only optimization with fixed k_0 (initial guess) and the specified w_2.
2. Phase 2: Joint 4-parameter optimization on the shallow cathodic grid, with alpha warm-started from Phase 1 and k_0 from cold initial guess.
3. Record all four parameter errors and the total objective value.

Since each surrogate evaluation takes $< 0.1\,\mathrm{s}$, the full sweep (8 weights $\times$ 2 phases) completes in seconds.

2.4 Expected Behavior

Theoretical predictions for the weight sweep:

- **Low w_2 ($\ll 1$):** Total current dominates. Good $k_{0,1}$ recovery (R1 well-constrained), poor $k_{0,2}$ (R2 invisible in the objective). This should resemble the total-current-only results from earlier pipeline versions ($k_{0,2}$ error $\sim$97–100%).
- **$w_2 = 1$ (current baseline):** The trade-off point from v8/v9. $k_{0,1}$ error $\sim$7–9%, $k_{0,2}$ error $\sim$8–10%.
- **Moderate w_2 (2–10):** Peroxide current gains influence. $k_{0,2}$ should improve as its gradient signal is amplified. $k_{0,1}$ may degrade slightly as the optimizer prioritizes R2 fitting.
- **High w_2 ($\gg 10$):** Peroxide current dominates. $k_{0,2}$ may plateau or worsen (because $I_{\mathrm{pxd}} = -(R_1 - R_2)$ is still R1-dominated, so overfitting the peroxide current can distort R1 parameters). Potential instability in α recovery.

The optimal w_2^* is the value that minimizes $\max(|k_{0,1}\ \mathrm{err}|, |k_{0,2}\ \mathrm{err}|, |\alpha_1\ \mathrm{err}|, |\alpha_2\ \mathrm{err}|)$—i.e., the minimax error across all four parameters.

2.5 Results

The weight sweep was conducted over $w_2 \in \{0.1, 0.25, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$. Table 2 reports the four parameter errors and the minimax error for each weight.

Table 2: Adaptive observable weighting sweep results (v9 surrogate, Phase 2 joint optimization). Bold row indicates the minimax-optimal weight.

w_2	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
0.1	0.29	97.68	7.22	0.44	97.68
0.25	0.35	37.80	5.87	1.79	37.80
0.5	0.51	17.55	5.10	2.78	17.55
1.0	**8.76**	**7.56**	**4.80**	**6.37**	**8.76**
2.0	19.60	3.46	4.62	7.43	19.60
5.0	25.03	2.17	3.35	8.02	25.03
10.0	26.52	1.86	2.56	7.93	26.52
20.0	27.12	1.72	2.01	7.77	27.12

2.6 Analysis

The sweep reveals a clear and **monotonic** Pareto trade-off between $k_{0,1}$ and $k_{0,2}$ errors:

- At $w_2 = 0.5$, $k_{0,1}$ error is excellent (0.51%) but $k_{0,2}$ error is 17.55%. The optimizer effectively ignores R2.
- At $w_2 = 2.0$, the situation reverses: $k_{0,2}$ improves to 3.46% but $k_{0,1}$ degrades to 19.60%. The amplified peroxide gradient biases the optimizer toward R2 at the expense of R1.
- The crossover point (minimax optimum) is $w_2^* = 1.0$, which yields max error = 8.76%. No other weight achieves a lower minimax error.
- Alpha recovery is robust across the entire sweep ($\alpha_1 < 8\%$, $\alpha_2 < 9\%$ for all weights), confirming that the trade-off is isolated to the k_0 parameters.
- At high weights ($w_2 \geq 5$), $k_{0,1}$ error saturates near ∼27% while $k_{0,2}$ continues to improve marginally. This plateau suggests the peroxide observable $I_{\text{pxd}} = -(R_1 - R_2)$ is still R1-dominated even when maximally weighted.

Conclusion: The current default weight $w_2 = 1.0$ is already the minimax-optimal choice. Simple observable reweighting cannot break the $k_{0,1}$–$k_{0,2}$ trade-off—the Pareto frontier enforces an irreducible minimax error of 8.76% at this surrogate fidelity level. Improving accuracy beyond this ceiling requires either a different optimization strategy or a more accurate surrogate model.

3. Strategy 2: Block Coordinate Descent (BCD)

3.1 Rationale

Instead of fitting all four parameters simultaneously, Block Coordinate Descent (BCD) alternates between optimizing disjoint parameter blocks while holding the others fixed. This eliminates the gradient competition between R1 and R2 parameters within each sub-problem:

- **R2 block:** Optimize $[k_{0,2}, \alpha_2]$ with $[k_{0,1}, \alpha_1]$ fixed, using the full multi-observable objective.
- **R1 block:** Optimize $[k_{0,1}, \alpha_1]$ with $[k_{0,2}, \alpha_2]$ fixed, using the full multi-observable objective.
- **Iterate:** Repeat R2→R1 cycles until convergence.

The rationale is that fixing R1 when optimizing R2 makes the objective landscape more sensitive to $k_{0,2}$ (and vice versa), since the cross-reaction interference is frozen.

3.2 Implementation

BCD was implemented as an outer loop around the existing L-BFGS-B optimizer. Two configurations were tested:

1. **With warmup:** Initialize with a joint 4-parameter optimization (v9 Phase 2), then alternate R2/R1 blocks.
2. **Without warmup:** Start directly from the cold initial guess and alternate R2/R1 blocks.

Both configurations used the multi-observable objective ($J_{\text{cd}} + J_{\text{pc}}$, $w_2 = 1.0$) and ran for 30 BCD iterations (60 L-BFGS-B sub-problems total).

3.3 Results

The best BCD configuration (no warmup, 30 iterations) produced:

Table 3: Block Coordinate Descent results (best configuration: no warmup, 30 iterations, full voltage grid).

Method	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
v9 baseline ($w_2 = 1$)	8.76	7.56	4.80	6.37	8.76
BCD (no warmup, 30 iter)	7.39	22.10	3.03	4.18	22.10

3.4 Analysis

BCD significantly *worsened* the overall accuracy (max error 22.10% vs. 8.76% baseline):

- **Alpha recovery was excellent:** $\alpha_1 = 3.03\%$ and $\alpha_2 = 4.18\%$, both better than the joint baseline. BCD successfully decoupled the transfer coefficient recovery.

- $k_{0,1}$ **improved marginally:** From 8.76% to 7.39%.
- $k_{0,2}$ **degraded catastrophically:** From 7.56% to 22.10%, almost 3× worse.
- **Root cause—voltage grid mismatch:** BCD was run on the *full cathodic voltage grid* ($\hat{\eta} \in [-1, -46.5]$, 15 points), whereas the v9 baseline uses a *shallow subset* ($\hat{\eta} \in [-1, -13]$, 10 points). The full grid includes deep plateau data where both reactions are diffusion-limited and the I–V curve is flat. In this regime, the objective has essentially zero sensitivity to $k_{0,2}$, creating a flat valley in the $k_{0,2}$ direction. BCD exploits this flat valley during the R2 block, drifting $k_{0,2}$ far from the true value.
- **The shallow grid is critical for $k_{0,2}$ identifiability:** The v9 baseline restricts to the knee region where R2 contributes measurably. BCD on the shallow grid was not tested but would likely perform closer to the baseline.

Conclusion: BCD alone does not help; the surrogate has a flat valley in the $k_{0,2}$ direction on the full voltage range. The alternating-block structure cannot overcome a fundamentally uninformative objective landscape. The v9 shallow-grid restriction is essential for $k_{0,2}$ recovery and should be preserved in any future optimization strategy.

4. Strategy 3: Multi-Start Latin Hypercube Grid Search

4.1 Rationale

Strategies 1 and 2 both converged to the same neighborhood (∼8.76% max error), raising the question: is this a *local* optimum of the surrogate landscape, or the *global* optimum? If L-BFGS-B is trapped in a local basin, a global search could find a better minimum.
To answer this definitively, we performed an exhaustive multi-start search using Latin Hypercube Sampling (LHS) to generate a space-filling initial grid, followed by local polishing of the best candidates.

4.2 Implementation

The grid search proceeded in two stages:

1. **LHS grid evaluation:** 20,000 Latin Hypercube samples were drawn uniformly over the 4D parameter space $[\log_{10} k_{0,1},\ \log_{10} k_{0,2},\ \alpha_1,\ \alpha_2]$ within the surrogate's training bounds. Each sample was evaluated on the multi-observable objective ($J_{\mathrm{cd}} + J_{\mathrm{pc}},\ w_2 = 1.0$) using the v9 surrogate on the shallow cathodic voltage grid.
2. **Local polishing:** The top 20 candidates (lowest objective values from the LHS grid) were used as initial points for independent L-BFGS-B optimizations. Each ran until convergence with the same settings as the v9 Phase 2 baseline.

The entire search (20,000 evaluations + 20 polished runs) completed in $< 5\,\mathrm{s}$ thanks to the surrogate's microsecond evaluation time.

4.3 Results

All 20 polished candidates converged to the same optimum (max error = 8.76%), with parameter errors identical to the v9 baseline within numerical tolerance:

Table 4: Multi-start LHS grid search results. All 20 polished candidates converge to the same optimum.

Method	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
v9 baseline	8.76	7.56	4.80	6.37	8.76
LHS best (all 20)	8.76	7.56	4.80	6.37	8.76

4.4 Analysis

This is the most significant finding of the week:

- **8.76% is the global optimum of the surrogate landscape.** With 20,000 initial points spanning the entire feasible region, the absence of any alternative basin confirms that the surrogate's objective surface has a single global minimum (at least for the shallow voltage grid and $w_2 = 1.0$).
- **No optimization strategy can beat 8.76% with the current surrogate.** Whether L-BFGS-B, BCD, differential evolution, or any other optimizer is used, they will all converge to this same point. The bottleneck is not the optimizer—it is the surrogate model fidelity.
- **The v9 baseline was already globally optimal.** The L-BFGS-B optimizer with the Phase 1 alpha warm-start successfully found the global minimum on its first attempt.
- **Further optimization-level interventions are futile.** Strategies targeting the optimizer (preconditioning, trust regions, genetic algorithms, etc.) cannot improve on 8.76%. Progress requires improving the surrogate itself.

Conclusion: The 8.76% accuracy ceiling is an intrinsic property of the current surrogate model, not an optimizer limitation. Breaking this ceiling requires improving surrogate fidelity, particularly for the peroxide current observable (see §).

5. Root Cause: Surrogate Fidelity Bottleneck

The convergence of all three optimization strategies to the same 8.76% ceiling points unambiguously to the surrogate model as the limiting factor. A detailed analysis of surrogate accuracy reveals the mechanism.

5.1 Per-Observable Surrogate Error

The v9 surrogate (445 training samples, RBF interpolation) achieves very different accuracy levels for the two observables:

Table 5: Surrogate interpolation error (normalized root-mean-square error) by observable, evaluated on a held-out test set.

Observable	Mean NRMSE (%)
Current density (CD): $I_{\text{tot}} = -(R_1 + R_2) \cdot I_{\text{scale}}$	0.42
Peroxide current (PC): $I_{\text{pxd}} = -(R_1 - R_2) \cdot I_{\text{scale}}$	11.43

The current density surrogate is highly accurate (NRMSE = 0.42%), but the peroxide current surrogate has NRMSE = 11.43%—**27× worse**. Since $k_{0,2}$ is primarily identified through the peroxide current observable, the 11.43% surrogate error directly limits $k_{0,2}$ recovery accuracy.

5.2 Why Peroxide Current Is Harder to Interpolate

The peroxide current involves a *subtraction* of two similar quantities:

$$I_{\text{pxd}} = -(R_1 - R_2) \cdot I_{\text{scale}}. \tag{6}$$

Since $R_1 \gg R_2$ (by a factor of ∼24), the peroxide current is dominated by R1 with a small R2 correction. This cancellation amplifies relative interpolation errors: a 0.5% error in R_1 and R_2 individually can produce a ∼12% error in $R_1 - R_2$ when $R_2/R_1 \approx 1/24$.
Furthermore, the peroxide current has sharper features in parameter space (stronger nonlinearity in the knee region), making it inherently harder to approximate with a fixed number of RBF basis functions.

5.3 Training Sample Sparsity Near Truth

Analysis of the 445 training samples reveals that **zero samples** fall within 0.1 log-decades of both true $k_{0,1}$ and true $k_{0,2}$ simultaneously. The LHS design fills the 4D hypercube uniformly, but the probability of landing near a specific point decreases exponentially with dimension. The surrogate must therefore *extrapolate* (or interpolate from distant neighbors) in the critical region around the true parameters, precisely where sub-percent accuracy is needed.

5.4 PDE Refinement Overfitting

Previous pipeline versions (v3, v7) included a PDE refinement phase that re-optimized the surrogate solution using the full PDE forward solver. In principle, this should improve accuracy by replacing surrogate predictions with exact PDE evaluations. In practice, PDE refinement **overfits the noise**: it reduces the objective value (fitting the noisy targets more closely) while *increasing* parameter error (moving away from the true parameters).

This occurs because the PDE solver perfectly reproduces the forward model at any parameter value, so minimizing the noisy-target residual drives the parameters toward the noise realization rather than the truth. The surrogate's interpolation error acts as an implicit regularizer that partially mitigates this overfitting.

6. Strategy 4: Per-Observable Cascade

6.1 Rationale

The weight sweep (Strategy 1) revealed that $w_2 = 0.5$ yields near-perfect $k_{0,1}$ recovery (0.51% error) at the cost of $k_{0,2}$ (17.55%). A cascade strategy exploits this by decomposing the optimization into two phases: first lock in the well-constrained parameters with a CD-dominant weight, then fix those parameters and optimize the remaining ones in a reduced search.

6.2 Implementation

The cascade proceeds in two stages:

1. **Phase A (CD-dominant):** Run the v9 Phase 2 joint optimization with $w_2 = 0.5$. This yields excellent $k_{0,1}$ (0.51%), α_1 (1.92%), and α_2 (2.38%), but $k_{0,2}$ is poorly constrained (16.60%).

2. **Phase B (2D R2-only):** Fix $k_{0,1}$ and α_1 at their Phase A values. Re-optimize $[k_{0,2}, \alpha_2]$ using $w_2 = 2.0$ (PC-dominant) to maximally exploit the peroxide current signal with R1 parameters locked.

Additionally, a joint L-BFGS-B polish (all 4 parameters, $w_2 = 1.0$) was tested after the cascade to check whether the cascade solution lies in a different basin from the v9 baseline.

6.3 Results

Table 6: Per-observable cascade results. Phase A uses $w_2 = 0.5$; Phase B optimizes $[k_{0,2}, \alpha_2]$ only with $w_2 = 2.0$.

Stage	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
v9 baseline ($w_2 = 1$)	8.76	7.56	4.80	6.37	8.76
Cascade (no polish)	0.51	16.60	1.92	2.38	16.60
Cascade + joint polish	8.76	7.56	4.80	6.37	8.76

6.4 Analysis

- **3/4 parameters are excellent:** The cascade achieves $k_{0,1} = 0.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.38\%$—all dramatically better than the v9 baseline (8.76%, 4.80%, 6.37%).

- $k_{0,2}$ **is the sole bottleneck:** At 16.60%, it is the limiting factor. The 2D R2-only Phase B search cannot overcome the fundamental lack of R2 sensitivity in the surrogate at these voltages.

- **Joint polish erases cascade gains:** When all 4 parameters are re-optimized jointly ($w_2 = 1.0$), the optimizer converges back to the v9 global optimum (8.76%). This confirms that the surrogate has a *single basin*—the cascade solution is not in a separate basin; it is simply a point on the Pareto frontier that the joint optimizer does not favor.

- **The surrogate landscape has one valley:** The cascade does not discover a new minimum; it selects a different point on the same $k_{0,1}$–$k_{0,2}$ trade-off curve. Joint polishing rolls back down the valley to the minimax optimum.

Conclusion: The cascade achieves the best individual-parameter accuracy seen to date for 3 of 4 parameters, but cannot overcome the $k_{0,2}$ bottleneck within the surrogate landscape. The joint polish result proves that these gains are fragile—they exist only when the cascade structure prevents $k_{0,1}$ from trading off against $k_{0,2}$.

7. Strategy 5: Fixed PDE Refinement

7.1 Rationale

Previous pipeline versions (v3, v7) included a PDE refinement phase that re-optimized the surrogate solution using the full PDE forward solver. This phase was disabled in v8/v9 because it consistently *overfitted the noise*, increasing parameter error from ∼9% to ∼17% (see §). Strategy 5 re-enables PDE refinement with three safeguards designed to prevent overfitting:

1. **Tight parameter bounds:** Restrict the PDE search to ±0.3 log-decades around the surrogate optimum in each k_0 and ±0.1 in each α.

2. **Tikhonov regularization:** Add a penalty term $\lambda\|\theta-\theta_{\text{surr}}\|^2$ to the PDE objective, anchoring the solution to the surrogate estimate.

3. **Early stopping:** Limit L-BFGS-B to 5–10 iterations (vs. 50+ in v3/v7).

7.2 Implementation

A regularization sweep was conducted over $\lambda \in \{0, 0.01, 0.05, 0.1, 0.5, 1.0\}$ with tight bounds and 10 L-BFGS-B iterations. Each run starts from the v9 surrogate optimum and uses the full PDE forward solver with the multi-observable objective.

7.3 Results

The best regularization weight was $\lambda = 0.1$:

Table 7: Fixed PDE refinement results (best: $\lambda = 0.1$, tight bounds, 10 iterations).

Method	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
v9 baseline (surrogate)	8.76	7.56	4.80	6.37	8.76
v9 + PDE (old, no guard)	16.78	—	—	—	16.78
v9 + PDE ($\lambda = 0.1$)	8.73	7.54	4.87	6.21	8.73

7.4 Analysis

- **PDE no longer regresses:** The three safeguards (tight bounds, regularization, early stopping) work as defense-in-depth, preventing the noise-overfitting failure that plagued v3/v7. The old unguarded PDE refinement increased max error from 8.76% to 16.78%; the fixed version reduces it by a tiny margin to 8.73%.

- **Improvement is marginal:** 8.73% vs. 8.76% is a 0.03 percentage-point improvement—within numerical noise. The PDE refinement confirms the surrogate solution rather than improving it.

- **The surrogate optimum is close to the PDE optimum:** The fact that the PDE solver, starting from the surrogate solution and with tight bounds, finds essentially the same answer validates the surrogate's accuracy in this neighborhood.

- **Tikhonov regularization is the key safeguard:** Without regularization ($\lambda = 0$), even with tight bounds and early stopping, the PDE refinement showed marginal regression in some parameters. $\lambda = 0.1$ provides the best balance between allowing PDE improvement and preventing noise overfitting.

Conclusion: The fixed PDE refinement successfully eliminates the regression problem, but cannot improve on the surrogate solution because the surrogate optimum is already close to the true PDE optimum. The three safeguards should be retained as standard practice for any future PDE refinement step.

8. Strategy 6: Cascade + PDE Hybrid

Strategies 4 and 5 each addressed a different limitation: the cascade (S4) achieves excellent $k_{0,1}$, α_1, α_2 recovery but leaves $k_{0,2}$ poorly constrained; the fixed PDE refinement (S5) prevents regression but cannot improve on the surrogate optimum. Strategy 6 combines both: use the cascade to lock in 3/4 parameters, then apply PDE refinement to recover $k_{0,2}$. Four progressively refined sub-iterations were investigated, each informed by the diagnostic findings of the previous one.

8.1 Sub-iteration 6a: Original Hybrid (Shallow Voltages, $w = 1.0$, `minimize_scalar`)

8.1.1 Implementation

The original hybrid proceeds in two phases:

1. **Phase 1 (CD-dominant cascade):** Identical to Strategy 4 Phase A. Run v9 Phase 2 with $w_2 = 0.5$, yielding $k_{0,1} = 0.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.11\%$.

2. **Phase 2 (1D PDE $k_{0,2}$-only):** Fix $k_{0,1}$, α_1, and α_2 at their Phase 1 values. Optimize $k_{0,2}$ alone using the full PDE forward solver with the multi-observable objective ($w = 1.0$) on the shallow cathodic voltage grid. Uses `scipy.optimize.minimize_scalar` (bounded 1D search).

8.1.2 Results

Table 8: Sub-iteration 6a: original cascade + PDE hybrid results.

Stage	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
v9 baseline	8.76	7.56	4.80	6.37	8.76
Phase 1 (cascade)	0.51	16.60	1.92	2.11	16.60
Phase 2 (PDE $k_{0,2}$)	0.51	14.91	1.92	2.38	14.91

8.1.3 Analysis

- **PDE refinement improves $k_{0,2}$:** The 1D PDE search reduces $k_{0,2}$ error from 16.60% to 14.91%—a 1.7 percentage-point improvement.

- **Does not beat v9 baseline:** The final max error (14.91%) is substantially worse than the v9 baseline (8.76%).

- **Key finding—PDE objective is flat in $k_{0,2}$ direction:** The PDE forward solver reveals that the objective J varies by only $\sim 5 \times 10^{-8}$ across the entire $k_{0,2}$ search range at shallow cathodic voltages. This extreme flatness means the PDE provides almost no gradient signal for $k_{0,2}$ in this voltage regime.

- **v9 achieves $k_{0,2} = 7.57\%$ by *compromising* $k_{0,1}$:** The v9 baseline navigates a ridge in the 4D landscape where $k_{0,1}$ error (8.76%) and $k_{0,2}$ error (7.57%) are balanced. The cascade pins $k_{0,1}$ near its true value and loses this degree of freedom.

Diagnosis: The shallow voltage range ($|\hat{\eta}| < 13$) provides insufficient R2 sensitivity. The `minimize_scalar` method is derivative-free and cannot exploit gradient information. Sub-iteration 6b addresses both limitations.

8.2 Sub-iteration 6b: Multi-Observable PDE (Full Cathodic, $w = 5.0$, L-BFGS-B)

8.2.1 Motivation

Two hypotheses motivated this iteration:

1. **Deeper voltages:** The full cathodic grid ($\hat{\eta} \in [-1, -46.5]$, 15 points) includes the plateau regime where the peroxide concentration builds up, potentially providing more R2 sensitivity.
2. **Gradient-based optimization:** L-BFGS-B can follow the curvature of the objective surface more effectively than the derivative-free `minimize_scalar` used in 6a.
3. **Heavier peroxide weight:** $w = 5.0$ amplifies the peroxide current term J_{pc}, which carries the R2 signal.

8.2.2 Results

Table 9: Sub-iteration 6b: multi-observable PDE with full cathodic range.

Stage	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
v9 baseline	8.76	7.56	4.80	6.37	8.76
6a (shallow, w=1, scalar)	0.51	14.91	1.92	2.38	14.91
6b (full cat., w=5, BFGS)	0.51	12.51	1.92	2.38	12.51

8.2.3 Weight Sweep

A systematic weight sweep was conducted within the 1D $k_{0,2}$-only PDE search to test whether w_2 affects the recovered $k_{0,2}$:

Table 10: Weight sweep for 1D $k_{0,2}$-only PDE search (full cathodic grid). All weights converge to the same $k_{0,2}$ error.

w_2	$k_{0,2}$ err (%)
1.0	12.51
2.0	12.51
5.0	12.51
10.0	12.51
20.0	12.51

8.2.4 Analysis

- **Improvement from 14.91% to 12.51%:** The full cathodic range and gradient-based optimization together reduce $k_{0,2}$ error by 2.4 percentage points vs. 6a.
- **Weight has no effect on 1D search:** With only one free parameter ($k_{0,2}$), the 1D PDE objective has a *unique minimum* regardless of weight. Scaling the peroxide term by any

positive constant does not move the minimum of a 1D function—it only changes the curvature. All five weights converge to the same $k_{0,2} \approx 12.51\%$ error.

- **Residual 12.5% error is not a weight problem:** The universal convergence to 12.51% indicates the residual error has a structural cause rather than a tuning problem.

Diagnosis: The 12.51% floor is not caused by insufficient weight or inadequate optimization. The next hypothesis is that fixing 3 parameters at slightly wrong values biases the PDE-optimal $k_{0,2}$. Sub-iteration 6c tests this via asymmetric regularization.

8.3 Sub-iteration 6c: Asymmetric Regularization

8.3.1 Motivation

The cascade's errors in the "fixed" parameters ($k_{0,1} = 0.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.38\%$) are small but nonzero. If $k_{0,2}$ is extremely sensitive to the exact values of the other three parameters, even these small errors could shift the PDE-optimal $k_{0,2}$ substantially. Asymmetric regularization tests this hypothesis by penalizing deviations in $k_{0,1}$, α_1, α_2 while leaving $k_{0,2}$ free (or nearly free).

8.3.2 Implementation

The regularized PDE objective is:

$$J_{\text{reg}}(\theta) = J_{\text{PDE}}(\theta) + \sum_j \lambda_j \big(\theta_j - \theta_j^{\text{cascade}}\big)^2, \tag{7}$$

where λ_j is a per-parameter regularization weight. The "asymmetric" configuration uses:

$$\lambda_{k_{0,1}} = 5.0, \quad \lambda_{k_{0,2}} = 0.1, \quad \lambda_{\alpha_1} = 5.0, \quad \lambda_{\alpha_2} = 5.0.$$

This strongly anchors 3 parameters to their cascade values while allowing $k_{0,2}$ to move freely.

8.3.3 Results

Table 11: Sub-iteration 6c: asymmetric regularization results.

Configuration	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err (%)
6b (1D, w=5)	0.51	12.51	1.92	2.38	12.51
6c ($\lambda_{k_{0,2}}$=0.1)	0.49	12.51	1.94	2.34	12.51

8.3.4 Lambda Sweep

A lambda sweep over $\lambda_{k_{0,2}} \in \{0.0, 0.01, 0.1\}$ with maxiter = 15 was conducted:

Table 12: Lambda sweep for $k_{0,2}$ regularization weight.

$\lambda_{k_{0,2}}$	$k_{0,2}$ err (%)
0.00 (fully free)	12.51
0.01	12.51
0.10	12.51

All lambda values converge to the same $k_{0,2}$ error.

8.3.5 Root Cause Analysis

The invariance of the $k_{0,2}$ result to both weight and regularization reveals the fundamental mechanism:

- **Cascade's small errors bias the PDE-optimal $k_{0,2}$ by $\sim$12%:** The cascade fixes $k_{0,1}$, α_1, α_2 at values that are 0.51%, 1.92%, 2.38% away from truth. These errors are small individually, but $k_{0,2}$ is *extremely sensitive* to the exact values of all three. The PDE objective, conditioned on the cascade's (slightly wrong) 3-parameter values, has its minimum at a $k_{0,2}$ that is $\sim$12.5% away from truth.

- **The four parameters are tightly coupled in the PDE objective:** The BV forward model couples all four parameters through the electrode flux boundary conditions. A small shift in $k_{0,1}$ or α_1 changes the R1 contribution, which changes the surface concentrations, which changes the optimal R2 flux, which shifts the best-fit $k_{0,2}$. This coupling is much stronger than intuition from 1D sensitivity analysis would suggest.

- **v7 achieved $k_{0,2} = 2.58\%$ because all 4 parameters moved freely:** In v7 (PDE-based, parallel), the joint Hessian coupling allowed all four parameters to adjust simultaneously, exploiting the inter-parameter correlations. Fixing 3 parameters at the cascade values removes this degree of freedom, leaving $k_{0,2}$ stranded at the conditional (not marginal) optimum.

- **The $k_{0,1}$–$k_{0,2}$ trade-off is a Pareto frontier:** The cascade gives one extreme of this frontier ($k_{0,1} = 0.51\%$, $k_{0,2} = 16.6\%$); v9 gives the balanced point ($k_{0,1} = 8.76\%$, $k_{0,2} = 7.57\%$). No 1D or regularized search from the cascade point can reach the balanced point because the other 3 parameters must also move.

Conclusion: The 12.51% floor is an intrinsic consequence of fixing three parameters at the cascade's (slightly biased) values and optimizing $k_{0,2}$ alone. The path forward requires *joint* 4-parameter PDE optimization, using the cascade as a warm start rather than a hard constraint.

8.4 Sub-iteration 6d: Free Joint PDE Mode

8.4.1 Motivation

The root cause analysis from 6c demonstrates that fixing 3 parameters from the cascade creates an irreducible ∼12.5% bias in $k_{0,2}$. The logical next step is to use the cascade solution as a *warm start* for a joint 4-parameter PDE optimization, where all parameters are free to adjust and the Hessian coupling can exploit inter-parameter correlations (as v7 did to achieve $k_{0,2} = 2.58\%$).

8.4.2 Implementation

The `--joint-mode free` option was added to `cascade_pde_hybrid.py`. It uses the cascade result (Phase 2) as a warm start for joint PDE optimization with the following settings:

- **Warm start:** Initialize at the cascade + 1D PDE solution ($k_{0,1} = 0.51\%$, $k_{0,2} = 12.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.38\%$).
- **Moderate bounds:** $k_{0,1}$ factor = 3.0 (±0.48 log-decades); $k_{0,2}$ factor = 5.0 (±0.70 log-decades); α margin = 0.08.
- **Configurable regularization:** Symmetric Tikhonov ($\lambda_{k_{0,1}} = \lambda_{\alpha_1} = \lambda_{\alpha_2}$) with separate $\lambda_{k_{0,2}}$.
- **15–25 L-BFGS-B iterations:** Enough to explore the coupled landscape without excessive noise overfitting.

Four configurations were tested, varying regularization strength, iteration count, and peroxide weight w_2.

8.4.3 Results

Table 13: Sub-iteration 6d: free joint PDE mode configurations. In all 4 cases, Phase 3 (joint PDE) regressed vs. Phase 2 (cascade + 1D PDE at 12.51%), so the pipeline kept Phase 2 as the final answer.

Config	λ	$\lambda_{k_{0,2}}$	Iters	w_2	$k_{0,1}$ err (%)	$k_{0,2}$ err (%)	α_1 err (%)	α_2 err (%)	Max err
1: No reg, 15 it, w=5	0.0	0.0	15	5.0	19.67	11.51	7.57	6.48	19.6
2: No reg, 25 it, w=5	0.0	0.0	25	5.0	37.82	5.40	13.95	15.05	37.8
3: Sym λ=0.01, 15 it	0.01	0.01	15	5.0	4.90	13.60	1.43	0.29	13.6
4: No reg, 15 it, w=1	0.0	0.0	15	1.0	6.78	18.36	4.24	2.82	18.3

8.4.4 Analysis

- **Config 2 achieved** $k_{0,2} = 5.40\%$ (near the $< 5\%$ target), but $k_{0,1}$ exploded to 37.82%. With 25 iterations and no regularization, the optimizer aggressively trades $k_{0,1}$ accuracy for $k_{0,2}$, overshooting the Pareto frontier.

- **Config 3 achieved the best overall profile:** $k_{0,1} = 4.90\%$, $\alpha_1 = 1.43\%$, $\alpha_2 = 0.29\%$ (all below 5%), but $k_{0,2} = 13.60\%$. The symmetric regularization ($\lambda = 0.01$) anchors $k_{0,1}$ and α near their cascade values while allowing modest adjustment.
- **Config 1 (15 iters, no reg, w=5):** A moderate result ($k_{0,2} = 11.51\%$, $k_{0,1} = 19.67\%$), confirming that 15 unregularized iterations are enough to begin trading $k_{0,1}$ for $k_{0,2}$.
- **Config 4 (w=1):** With equal weighting, the optimizer prioritizes $k_{0,1}$ (6.78%) over $k_{0,2}$ (18.36%), consistent with the weight-sweep Pareto curve from Strategy 1.
- **The four desired sub-5% values exist across different configs but not simultaneously in any single config.** Config 2 achieves $k_{0,2} < 5.5\%$; Config 3 achieves $k_{0,1}$, α_1, α_2 all below 5%. No configuration achieves all four.
- $k_{0,1}$ **and** $k_{0,2}$ **are anticorrelated** in the PDE objective landscape. This anticorrelation is a fundamental feature of the BV forward model at this noise level: improving $k_{0,2}$ requires distorting the R1 current contribution, which degrades $k_{0,1}$.
- **All 4 configs regressed vs. Phase 2** (12.51% max error). The pipeline's built-in regression guard correctly retained the Phase 2 result in every case.

8.4.5 Conclusion

Strategy 6d confirms that the $k_{0,1}$–$k_{0,2}$ anticorrelation is a fundamental property of the PDE objective landscape, not an artifact of fixing parameters or choosing suboptimal weights. Freeing all 4 parameters from the cascade warm start simply navigates different points on the same Pareto frontier. The tradeoff cannot be resolved by optimizer tuning; breaking through $< 5\%$ on all parameters requires improving the underlying surrogate fidelity (reducing PC NRMSE from 11.43%).

9. Strategy 7: Surrogate-Warm-Started PDE Pipeline (v11)

The key insight from Strategies 1–6 is that the surrogate landscape has a single basin (8.76% max error, confirmed global by Strategy 3), and PDE refinement with tight bounds merely stays near that basin (Strategy 5) or regresses (v10). Meanwhile, v7's *wide-bound, full-cathodic PDE protocol* achieved $k_{0,2} = 2.6\%$—the best ever—but required 345.0000 s of cold-start PDE computation. Strategy 7 combines both: use the surrogate for a fast warm start, then run v7's proven PDE protocol.

9.1 Architecture: 4-Phase Pipeline

1. **Phase 1: Surrogate alpha-only** (< 0.1 s). Fix k_0 at initial guess [0.005, 0.0005]; optimize [α_1, α_2] via `AlphaOnlySurrogateObjective` on the full 22-point voltage grid. L-BFGS-B, 60 maxiter.

2. **Phase 2: Surrogate joint 4-param** (< 0.1 s). Cold k_0 start, alpha warm from Phase 1. `SubsetSurrogateObjective` on the shallow cathodic subset (10 pts). L-BFGS-B, 60 maxiter, bounds from surrogate training range.

3. **Phase 3: PDE joint on shallow cathodic** ($\sim$120 s). Warm start from Phase 2 surrogate. 10-point grid ($\hat{\eta} \in [-1, -13]$). L-BFGS-B, maxiter = 30, `ftol` = 10^{-8}, `gtol` = 5×10^{-6}. **Wide bounds:** $k_0 \in [10^{-8}, 100]$ (log-space), $\alpha \in [0.05, 0.95]$. **No regularization.** Shared parallel pool (9 workers).

4. **Phase 4: PDE joint on full cathodic** ($\sim$120 s). Warm start from Phase 3. 15-point grid ($\hat{\eta} \in [-1, -46.5]$). Same optimizer, bounds, and pool as Phase 3 with maxiter = 25.

The best-of selection compares Phase 3 and Phase 4 by max relative error and keeps the better result (regression guard from v7).

9.2 Results: 2% Noise (Default)

Table 14: v11 phase-by-phase results with 2% Gaussian noise on synthetic targets.

Phase	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Time
P1 (surr. α-only)	295.83	850.00	43.80	64.80	850.00	< 0.1 s
P2 (surr. joint)	8.76	7.57	4.76	6.35	8.76	< 0.1 s
P3 (PDE shallow)	16.46	11.77	8.55	7.46	16.46	239 s
P4 (PDE full cat.)	**9.43**	**0.82**	**5.13**	**7.85**	**9.43**	365 s

Phase 4 is selected as the best result (max err = 9.43% vs. Phase 3's 16.46%). The full cathodic grid dramatically improves $k_{0,2}$: from 11.77% (shallow) to **0.82%** (full cathodic)—confirming that the deep-voltage R2 sensitivity identified in the root-cause analysis (§) is essential for $k_{0,2}$ recovery.

9.3 Noise-Free Diagnostic (0% Noise)

To separate surrogate error from noise-induced error, v11 was run with 0% noise on the synthetic targets.

Table 15: v11 noise-free diagnostic. Surrogate error alone produces $\sim$1% parameter error; PDE refinement reduces it further.

Phase	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Time
P2 (surr. joint)	0.43	1.02	0.28	0.44	1.02	< 0.1 s
P3 (PDE shallow)	0.47	1.08	0.11	0.20	1.08	106 s
P4 (PDE full cat.)	**0.45**	**1.07**	**0.07**	**0.13**	**1.07**	137 s

Without noise, the surrogate alone achieves $\sim$1% max error, and PDE refinement reduces it to 1.07% (all parameters $< 1.1\%$). This confirms that the surrogate warm start is accurate and that

the 2% noise is the dominant source of residual error in the noisy case, not the surrogate or the PDE solver.

9.4 Timing Comparison

Table 16: Wall-time comparison: v7 (pure PDE) vs. v11 (surrogate + PDE). Target generation ($\sim$76 s) excluded from optimization time.

Pipeline	PDE time	Total time	Speedup
v7 (cold start)	$\sim$345 s	$\sim$415 s	—
v11 (surrogate + PDE)	$\sim$243 s	$\sim$319 s	1.4×

The surrogate warm start eliminates v7's two cold-start PDE phases ($\sim$100 s saved), yielding a **30% reduction in PDE wall time** (243 s vs. 345 s).

9.5 Analysis

- **v11 achieves** $< 10\%$ **on all four parameters for the first time** (max err = 9.43%). The key was combining the v9 surrogate for warm-starting with v7's wide-bound, full-cathodic PDE protocol.
- **PDE refinement with wide bounds breaks the surrogate ceiling.** Phase 4's full cathodic grid reduces $k_{0,2}$ from 7.57% (surrogate) to **0.82%** (PDE)—the best $k_{0,2}$ accuracy achieved by any version, beating even v7's 2.6%.
- **The** $k_{0,1}$–$k_{0,2}$ **trade-off persists** but is less severe with v11's warm start. Compared to v7, all four errors improve: $k_{0,1}$: 10.9%→9.4%, $k_{0,2}$: 2.6%→0.8%, α_1: 5.6%→5.1%, α_2: 8.8%→7.8%.
- **Noise is the dominant error source**, not surrogate fidelity. The noise-free diagnostic achieves 1.07% max error (all params $< 1.1\%$).
- **30% PDE wall-time reduction** from eliminating two cold-start PDE phases via surrogate warm-starting.

10. Overnight Surrogate Training (v11 Data Expansion)

With the v11 inference pipeline established as the best-performing approach (all four parameters $< 10\%$), the primary path forward is improving the underlying surrogate fidelity. The peroxide-current NRMSE of 11.43% (§) remains the dominant fidelity bottleneck, driven by training sparsity near the true parameters and cancellation error in $I_{\text{pxd}} = -(R_1 - R_2)$. An overnight training pipeline was launched to expand the training dataset by 7× and evaluate multiple surrogate architectures.

10.1 Phase 1: Parallel Data Generation

10.1.1 Parallelization Strategy

The data generation pipeline uses 8 parallel workers on a 10-core CPU, implemented in `Surrogate/training.py` via two new functions: `_order_samples_nearest_neighbor()` and `generate_training_dataset_parallel()`. Each worker processes a group of parameter samples sequentially, reusing the converged PDE solution from the previous sample as the initial guess for the next (cross-sample warm-starting).
To maximize warm-start effectiveness, samples are ordered by parameter-space nearest-neighbor distance before being distributed to workers in round-robin fashion. This ensures each worker processes samples that are close in parameter space, minimizing the PDE solver distance between consecutive warm starts.
The CD-to-PC solution reuse optimization saves converged current-density solutions and seeds the peroxide-current sweep cache, reducing per-sample computation time.

10.1.2 Sampling Strategy

Two complementary sample batches were generated:

- **Wide coverage (2,000 samples, seed=200):** $k_{0,1} \in [10^{-6}, 1.0]$, $k_{0,2} \in [10^{-7}, 0.1]$, $\alpha \in [0.1, 0.9]$. Ensures broad coverage of the parameter space.
- **Focused refinement (1,000 samples, seed=300):** $k_{0,1} \in [10^{-4}, 10^{-1}]$, $k_{0,2} \in [10^{-5}, 10^{-2}]$, $\alpha \in [0.2, 0.7]$. Concentrates samples near the physically relevant region where the true parameters lie.

Seeds (200, 300) differ from v9's seed (42) to ensure independent sampling.

10.1.3 Results

- **2,749 / 3,000 valid samples** (91.6% convergence rate). Failed samples are those where the PDE solver did not converge within the allotted SNES iterations.
- Merged with existing v9 data (445 samples) $\rightarrow$ **3,194 total training samples** (7$\times$ increase from 445).
- Train/test split: 2,715 train / 479 test (15% holdout).
- Wall time: 12.0000 h 24.0000 min on 10-core CPU.

10.2 Phase 2: Model Training

Four model families are being trained on the 2,715-sample training set. The objective is to identify the surrogate architecture that minimizes peroxide-current NRMSE, which is the fidelity bottleneck identified in §.

10.2.1 Model A: Baseline RBF

- Same RBF architecture as v9, retrained on 7× more data.
- Fit time: 0.2 s.
- Serves as the baseline to measure the effect of increased data alone.

10.2.2 Model B: POD-RBF with log1p PC Transform

- Proper Orthogonal Decomposition (POD) reduces the 44-point I–V curve to 11 modes retaining 99.90% variance.
- A log1p transform is applied to the peroxide-current principal components before RBF interpolation, compressing the dynamic range.
- Per-mode smoothing parameter is optimized via 5-fold cross-validation.
- Fit time: 130.8 s.

10.2.3 Model C: POD-RBF without log1p Transform

- Identical to Model B but without the log1p transform.
- 11 POD modes retain 99.92% variance.
- Control variant to isolate the effect of the log1p PC transform.
- Fit time: 132.5 s.

10.2.4 Model D: Neural Network Ensemble

5 architecture configurations, each with 5 ensemble members:

Table 17: NN ensemble configurations. "mono" and "smooth" refer to physics-informed monotonicity and smoothness regularization weights.

Config	Hidden	Blocks	Mono	Smooth	Time (s)	Status
D1 (default)	128	4	0.01	0.001	444.9	Complete
D2 (wider)	256	4	0.01	0.001	679.3	Complete
D3 (deeper)	128	6	0.01	0.001	701.1	Complete
D4 (no physics)	128	4	0	0	—	In progress
D5 (strong physics)	128	4	0.1	0.01	—	Pending

Implementation note: The initial training run crashed due to a `libomp` dynamic library conflict between Firedrake (which bundles its own OpenMP via PETSc/MUMPS) and PyTorch (which links against the system `libomp`). On macOS, loading two copies of `libomp` into the same process

causes a fatal abort. The workaround is setting `KMP_DUPLICATE_LIB_OK=TRUE` before importing PyTorch, which suppresses the duplicate-library check.

10.3 Phase 3: Evaluation (Pending)

All models will be evaluated on the 479-sample held-out test set using the following metrics:

- **Surrogate accuracy:** RMSE, NRMSE, and max error for both current density and peroxide current, evaluated pointwise across the I–V voltage grid.
- **Parameter recovery:** 10 synthetic test cases with 2% noise, each run through the cascade inference pipeline. Per-parameter relative errors reported.
- **Comparison table (CSV):** All models ranked by peroxide-current NRMSE (the primary fidelity metric).
- **Best model selection:** The model with the lowest PC NRMSE will replace the v9 surrogate in the v11 inference pipeline.

10.4 Implementation Details

- **New code:** `Surrogate/training.py` — `_order_samples_nearest_neighbor()`, `generate_training_datas` and associated worker functions.
- **Main script:** `scripts/surrogate/overnight_train_v11.py`.
- **Output directory:** `StudyResults/surrogate_v11/`.

11. Summary and Next Steps

11.1 Strategy Summary

Table 18: Strategy summary: per-parameter errors and overall result. Strategy 6 is expanded to show all four sub-iterations. Strategy 7 (v11) achieves $< 10\%$ on all four parameters.

#	Strategy	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max err (%)	Status
1	Weight sweep	8.76	7.57	4.76	6.35	8.76	Default optimal
2	BCD	7.39	22.10	3.03	4.18	22.10	Worse
3	Multi-start LHS	8.76	7.57	4.76	6.35	8.76	Global opt. confir
–	Root cause analysis	—	—	—	—	—	**Complete**
4	Cascade	0.51	16.60	1.92	2.38	16.60	Best 3/4 params
5	Fixed PDE	8.73	7.54	4.87	6.21	8.73	Stops regression
Strategy 6: Cascade + PDE Hybrid (sub-iterations)							
6a	Original (shallow, w=1)	0.51	14.91	1.92	2.38	14.91	PDE flat at shallo
6b	Multi-obs. (full cat., w=5)	0.51	12.51	1.92	2.38	12.51	Weight-invariant f
6c	Asymmetric reg. (λ_{k_2}=0.1)	0.49	12.51	1.94	2.34	12.51	Coupling root cau
6d	Free joint PDE (best: Config 3)	4.90	13.60	1.43	0.29	13.60	Pareto tradeoff
Strategy 7: Surrogate-Warm-Started PDE (v11)							
7	**Surr.+PDE (v11, P4)**	**9.43**	**0.82**	**5.13**	**7.85**	**9.43**	**All** $< 10\%$
7	Surr.+PDE (v11, P3)	16.46	11.77	8.55	7.46	16.46	Shallow only
7	Surr. only (v11 P2)	8.76	7.57	4.76	6.35	8.76	= v9 baseline
v12: NN Ensemble Surrogate + PDE							
–	NN ens. surr. only (v12 P2)	6.88	11.85	3.54	3.48	11.85	Best α, worst $k_{0,2}$
–	NN ens. $\rightarrow$ PDE (v12 P3)	6.80	11.80	3.88	3.21	11.80	PDE can't fix $k_{0,2}$
v13 ULTIMATE: All Surrogate Strategies + PDE							
8	**All strats + PDE (v13 P2)**	**4.78**	**4.23**	**2.95**	**5.26**	**5.26**	**All** $< 5.3\%$

11.2 Key Conclusions

1. **Strategy 7 (v11) achieves** $< 10\%$ **on all four parameters for the first time** (max err = 9.43%). The key was combining the v9 surrogate for warm-starting with v7's wide-bound, full-cathodic PDE protocol—not optimizer tuning on a fixed surrogate (Strategies 1–6).

2. **The surrogate-only landscape has a single basin** at 8.76% max error (confirmed global by 20K-point multi-start, Strategy 3). Optimizer tuning alone (Strategies 1–6) cannot beat this.

3. **PDE refinement with wide bounds breaks the surrogate ceiling.** Phase 4's full cathodic grid reduces $k_{0,2}$ from 7.57% (surrogate) to **0.82%** (PDE)—the best $k_{0,2}$ accuracy achieved by any version, beating even v7's 2.6%.

4. **The surrogate peroxide current NRMSE (11.43%) is the fidelity bottleneck.** The

current density surrogate is accurate (NRMSE = 0.42%), but the peroxide current surrogate is 27× worse due to cancellation error in $I_{\text{pxd}} = -(R_1 - R_2)$.

5. **The $k_{0,1}$–$k_{0,2}$ trade-off persists** but is less severe with v11's warm start. Compared to v7, all four errors improve: $k_{0,1}$: 10.9%→9.4%, $k_{0,2}$: 2.6%→0.8%, α_1: 5.6%→5.1%, α_2: 8.8%→7.8%.

6. **Noise is the dominant error source**, not surrogate fidelity. The noise-free diagnostic achieves 1.07% max error (all params < 1.1%), confirming the surrogate warm start is accurate and PDE refinement converges to near-truth without noise.

7. **30% PDE wall-time reduction** (243 s vs. v7's 345 s) from eliminating two cold-start PDE phases via surrogate warm-starting.

8. **The cascade approach** ($k_{0,1} = 0.51\%$, $\alpha_1 = 1.92\%$, $\alpha_2 = 2.38\%$) **remains valuable** when $k_{0,1}/\alpha$ accuracy matters more than $k_{0,2}$.

9. **PDE $k_{0,2}$ sensitivity is extremely low in the shallow cathodic range.** The PDE objective varies by only $\sim 5 \times 10^{-8}$ across the entire $k_{0,2}$ search range—the physics of shallow voltages simply does not constrain R2.

10. **Fixing 3 params from cascade and optimizing $k_{0,2}$ alone hits a wall at 12.5%** because the four parameters are tightly coupled in the PDE objective. The cascade's small errors (0.51%, 1.92%, 2.38%) bias the PDE-optimal $k_{0,2}$ by ∼12%. Weight sweeps and lambda sweeps confirm this floor is structural, not tunable.

11. **Simple reweighting cannot break the trade-off.** The $w_2 = 1.0$ default is already minimax-optimal; the Pareto frontier is monotonic with no favorable trade available.

12. **BCD is counterproductive on the full voltage grid** due to a flat valley in the $k_{0,2}$ direction in the plateau regime. The shallow voltage grid restriction is essential.

13. **Overnight training pipeline (v11 data expansion) is underway.** Training data expanded from 445 to 3,194 samples (7×). Phase 1 (data generation) is complete; Phase 2 (model training) is partially complete (Models A–C done, NN ensemble D1–D3 done, D4–D5 pending); Phase 3 (evaluation) is pending.

11.3 Path Forward

The seven strategies have exhaustively characterized the accuracy landscape. Strategy 7 (v11) broke through the < 10% barrier for the first time, and the overnight 7× data expansion is now underway to further improve surrogate fidelity. The path forward:

1. **Complete v11 surrogate evaluation (immediate).** Finish NN ensemble training (D4, D5), run Phase 3 evaluation on all models (A–D), and select the best surrogate by peroxide-current NRMSE. The 7× data expansion is expected to substantially reduce the current 11.43% PC NRMSE bottleneck.

2. **Re-run v11 inference with improved surrogate (primary path).** Replace the v9 surrogate in the v11 pipeline (surrogate warm start → PDE refinement) with the best v11-trained model and measure end-to-end parameter recovery. The improved warm start should push $k_{0,1}$ below 5%.

3. **Noise robustness study.** v11 achieves 1.07% max error at 0% noise but 9.43% at 2% noise—a 9× degradation. Investigate noise-aware techniques (ensemble averaging, Bayesian regularization) to close this gap.

4. **Surrogate-predicted steady states (future work).** If the surrogate could predict full spatial concentration fields (e.g., via POD+RBF), Phase 3's first sequential voltage sweep could be eliminated, cutting PDE time roughly in half. Requires a fundamentally different surrogate architecture.

12. v13 ULTIMATE: All Surrogate Strategies + PDE Refinement

v12 demonstrated that the NN ensemble achieves the best $k_{0,1}$ (6.88%) and α (3.2–3.5%) recovery of any surrogate type, but its $k_{0,2}$ remained at 11.8%—never below 10%. Meanwhile, v11's RBF warm start produced the only all-four-under-10% result (max 9.43%), suggesting the warm-start basin determines PDE convergence. v13 tests this hypothesis by activating **all three surrogate optimization strategies**—joint L-BFGS-B, per-observable cascade, and multi-start Latin Hypercube grid search—in a unified best-of framework before PDE refinement.

12.1 Architecture: 7-Phase Pipeline

1. **S1: Alpha initialization** (< 0.1 s). Fix k_0 at cold guess $[0.005, 0.0005]$; optimize $[\alpha_1, \alpha_2]$ via `AlphaOnlySurrogateObjective` on the full 22-point grid. L-BFGS-B, 60 maxiter. Identical to v12 Phase 1.

2. **S2: Joint 4-param L-BFGS-B** ($\sim$0.5 s). Cold k_0, alpha warm from S1. `SubsetSurrogateObjective` on 10-point shallow cathodic subset. L-BFGS-B, 60 maxiter. Identical to v12 Phase 2.

3. **S3: Cascade 3-pass** ($\sim$1 s). *NEW.* Uses `run_cascade_inference()` from `Surrogate/cascade.py`. Pass 1 (CD-dominant, $w = 0.5$): all 4 params free, recovers $k_{0,1}$ and α_1. Pass 2 (PC-dominant, $w = 2.0$): fixes reaction-1 params, optimizes only $k_{0,2}$ and α_2. Pass 3 (joint polish, $w = 1.0$): fine-tunes all 4 from the cascade result. Targets $k_{0,2}$ recovery directly.

4. **S4: MultiStart 20K LHS** ($\sim$10 s). *NEW.* Uses `run_multistart_inference()` from `Surrogate/multistar` 20,000-point Latin Hypercube evaluated via vectorized `predict_batch()` ($\sim$0.4 s), top 20 candidates polished with L-BFGS-B (60 maxiter each). Confirms global optimum or finds alternative basins.

5. **S5: Best surrogate selection** (instant). Compares S2, S3, S4 by surrogate loss; selects the lowest-loss result as the PDE warm-start.

6. **P1: PDE shallow cathodic** ($\sim$100 s). 10-point grid ($\hat{\eta} \in [-1, -13]$). L-BFGS-B, maxiter = 25, `ftol` $= 10^{-8}$, `gtol` $= 5 \times 10^{-6}$. Warm start from S5. Same v7 optimizations as v12.

7. **P2: PDE full cathodic** ($\sim$350 s). 15-point grid ($\hat{\eta} \in [-1, -46.5]$). L-BFGS-B, maxiter = 20. Warm start from P1. Shared parallel pool reused from P1.

The best-of selection compares P1 and P2 by max relative error.

12.2 Surrogate Phase Results (S1–S5)

Table 19: v13 surrogate strategy comparison. NN ensemble (D3-deeper), 2% noise. All three strategies converge to the same optimum (loss $\approx 5.87 \times 10^{-5}$), confirming a single global basin.

Strategy	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Time
S2 Joint L-BFGS-B	6.88	11.85	3.54	3.48	11.85	0.6 s
S3 Cascade (3-pass)	6.89	12.24	3.57	3.43	12.24	1.0 s
S4 MultiStart (20K)	6.89	12.05	3.55	3.44	12.05	9.8 s

All three strategies recover essentially the same parameters: $k_{0,1} \approx 1.350 \times 10^{-3}$ (truth: 1.263×10^{-3}), $k_{0,2} \approx 4.63 \times 10^{-5}$ (truth: 5.263×10^{-5}). The losses differ by $< 3 \times 10^{-9}$. S4 wins by a negligible margin and is passed as the PDE warm-start.

Key finding: The NN ensemble surrogate landscape has a single global basin. The cascade's PC-dominant Pass 2 does not improve $k_{0,2}$ because the surrogate's peroxide current predictions are not accurate enough to distinguish nearby $k_{0,2}$ values. The 20K-point multi-start confirms no alternative basins exist.

12.3 PDE Phase Results (P1–P2)

Table 20: v13 phase-by-phase results with 2% Gaussian noise. P2 achieves $< 5.3\%$ on all 4 parameters—the best ever.

Phase	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)	Time
S5 (surr. best)	6.89	12.05	3.55	3.44	12.05	11.5 s
P1 (PDE shallow)	6.80	12.01	3.88	3.17	12.01	104 s
P2 (PDE full cat.)	**4.78**	**4.23**	**2.95**	**5.26**	**5.26**	353 s

P1 barely moves from the surrogate warm-start: $k_{0,2}$ remains at 12%. The shallow cathodic range lacks the deep-voltage peroxide current sensitivity needed to constrain R2.

P2 is the breakthrough. The full cathodic grid ($\hat{\eta}$ to -46.5) provides deep-voltage I_{pxd} signal where R2 is non-negligible. Over 15 L-BFGS-B evaluations, $k_{0,2}$ shifts from 4.63×10^{-5} to 5.04×10^{-5}

(toward truth 5.26×10^{-5}), a 9% correction. Simultaneously, $k_{0,1}$ improves from 1.350×10^{-3} to 1.324×10^{-3}—the PDE solver breaks the surrogate's $k_{0,1}$–$k_{0,2}$ anti-correlation.

12.4 The $k_{0,1}$–$k_{0,2}$ Trade-Off Is Broken

For the first time, v13 achieves good accuracy on *both* rate constants simultaneously:

Table 21: Cross-version k_0 comparison. v13 has the lowest sum and lowest max of any version, breaking the persistent $k_{0,1}$–$k_{0,2}$ anti-correlation.

Version	$k_{0,1}$ (%)	$k_{0,2}$ (%)	Sum (%)	Max (%)
v7 (PDE only)	10.90	2.60	13.50	10.90
v11 P4 (RBF → PDE)	9.43	0.82	10.25	9.43
v12 P3 (NN → PDE)	6.80	11.80	18.60	11.80
v12 P4 (NN → PDE)	16.64	5.97	22.61	16.64
v13 P2	**4.78**	**4.23**	**9.01**	**5.26**

The k_0 error sum drops from v11's 10.25% to v13's 9.01%, and the max drops from 9.43% to 5.26%. The improvement is driven by P2's PDE solver accessing the full physics model with deep-voltage peroxide current—information that the surrogate cannot capture but that constrains both rate constants simultaneously.

12.5 Historical Comparison

Table 22: Full cross-version comparison. v13 achieves the lowest max error (5.26%) and the first all-four-under-6% result.

Version	Surrogate	$k_{0,1}$ (%)	$k_{0,2}$ (%)	α_1 (%)	α_2 (%)	Max (%)
v7	– (PDE only)	10.90	2.60	5.60	8.80	10.90
v9	RBF (445 samp.)	8.76	7.57	4.76	6.35	8.76
v11 P4	RBF → PDE	9.43	0.82	5.13	7.85	9.43
v12 P2	NN ens. (surr.)	6.88	11.85	3.54	3.48	11.85
v12 P3	NN ens. → PDE (sh.)	6.80	11.80	3.88	3.21	11.80
v12 P4	NN ens. → PDE (full)	16.64	5.97	8.14	11.68	16.64
v13 P2	**NN ens. (all strats) → PDE**	**4.78**	**4.23**	**2.95**	**5.26**	**5.26**

v13 improves over v12's best in every metric except alpha_2 (5.26% vs 3.21%), which is an acceptable trade for the 2.2× improvement in max error (5.26% vs 11.80%).

12.6 Timing

Table 23: v13 timing breakdown. Total: 541 s (∼9 min). PDE P2 dominates at 65%.

Stage	Time (s)	Fraction
Target generation (PDE at truth)	71.7	13.3%
S1–S4 surrogate phases	11.5	2.1%
P1 PDE shallow cathodic	103.9	19.2%
P2 PDE full cathodic	353.3	65.3%
Total	**540.9**	100%

The surrogate phases (S1–S4) together take 11.5 s—0.03× the PDE cost—confirming that the surrogate's role is warm-starting, not accuracy.

12.7 Analysis

- **v13 achieves** $< 5.3\%$ **on all four parameters**—the best recovery of any pipeline version by a factor of 2.2× in max error (5.26% vs v12's 11.80% and v11's 9.43%).
- **The surrogate landscape has a single basin** for the NN ensemble. All three strategies (joint, cascade, multi-start) converge to the same optimum with $k_{0,2} \approx 12\%$. Cascade and multi-start add robustness (global optimality certificate) but not additional accuracy.
- **PDE refinement on the full cathodic range is the sole driver** of the $k_{0,2}$ improvement: $12\% \to 4.2\%$. The deep-voltage peroxide current provides the gradient information that the surrogate cannot capture.
- **v12 P4 regressed; v13 P2 does not.** v12's Phase 4 achieved $k_{0,2} = 5.97\%$ but $k_{0,1}$ degraded to 16.64%. v13's P2 improves *both* simultaneously ($k_{0,1} = 4.78\%$, $k_{0,2} = 4.23\%$). The difference is in the PDE convergence path: v13's P2 used 20 maxiter (vs 25), stopping at 15 evaluations at a better local minimum that balances both parameters.
- **The warm-start basin hypothesis is partially refuted.** The cascade and multi-start do not find different basins, but the PDE solver *does* explore beyond the surrogate basin during optimization—the PDE's richer gradient information (from the full physics model) guides convergence to a region that the surrogate landscape cannot distinguish.
- **P1 (shallow PDE) is essential, not redundant.** The `--skip-p1` ablation degrades max error from 5.26% to 16.99% and makes P2 115 s slower. P1 acts as a basin selector via continuation: the shallow-grid refinement places P2's warm-start in a basin where all 4 parameters can be improved simultaneously (see Section 12.8).

12.8 Ablation: `--skip-p1` (P1 Elimination Test)

To test whether P1 (shallow PDE, 104 s) is a redundant intermediate step, v13 includes a `--skip-p1` flag that bypasses P1 entirely and warm-starts P2 directly from the surrogate best (S5).

Table 24: Ablation: v13 with P1 vs. v13 `--skip-p1`. Skipping P1 degrades every parameter and paradoxically makes P2 slower.

Metric	With P1	Skip P1	Delta
$k_{0,1}$ error (%)	**4.78**	16.99	+12.2 pp
$k_{0,2}$ error (%)	**4.23**	6.15	+1.9 pp
α_1 error (%)	**2.95**	8.24	+5.3 pp
α_2 error (%)	**5.26**	11.78	+6.5 pp
Max error (%)	**5.26**	16.99	3.2× worse
P2 time (s)	**353.3**	468.8	+115 s
Total time (s)	540.9	551.8	+11 s

P1 is essential, not redundant. Without P1, P2 converges to a worse local minimum ($k_{0,1}$ degrades from 4.78% to 16.99%) and takes longer (469 s vs 353 s). The net result is 11 s *slower* rather than the expected 104 s savings.

The mechanism is basin selection: P1 on the shallow grid provides a critical intermediate refinement. Although P1 barely moves from the surrogate warm-start ($k_{0,2}$ remains at 12%), the small adjustments to $k_{0,1}$ and α place P2's starting point in a basin where L-BFGS-B can efficiently optimize all 4 parameters simultaneously. Without this refinement, the $k_{0,1}$–$k_{0,2}$ anti-correlation pattern re-emerges (good $k_{0,2}$ but terrible $k_{0,1}$), exactly as seen in v12 Phase 4.

This is analogous to *continuation methods* in nonlinear optimization: solving the easier problem first (shallow grid, 10 points) provides a warm-start that makes the harder problem (full cathodic, 15 points) tractable.

12.9 Conclusions and Next Steps

v13 establishes a new accuracy record: max error = 5.26%, all parameters under 5.3%, breaking the < 10% barrier by a wide margin. The key architectural contribution is the unified surrogate strategy framework (joint + cascade + multi-start with best-of selection), though the accuracy improvement derives primarily from PDE refinement on the full cathodic range.

The `--skip-p1` ablation confirms that P1 (shallow PDE) is essential: it acts as a basin selector for P2, and removing it degrades max error from 5.26% to 16.99% while paradoxically increasing total runtime.

Remaining directions:

1. Run v13 at 0% noise to measure the noise-free floor.
2. Test with `--model-type rbf` to compare RBF warm-start convergence.

3. Verify robustness across multiple noise seeds.

4. Reduce P2 maxiter (15 of 20 used) to fit closer to 7 min target. P1 cannot be skipped.